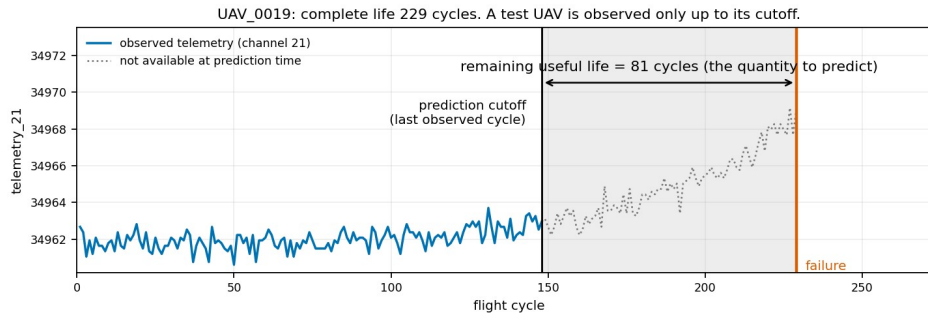


Estimating remaining useful life from a partial UAV telemetry history



How we reached a public R^2 of 0.87652

Joest
Homann

Five submissions, five decisions

Each point is a submitted model; the label names the decision that run introduced

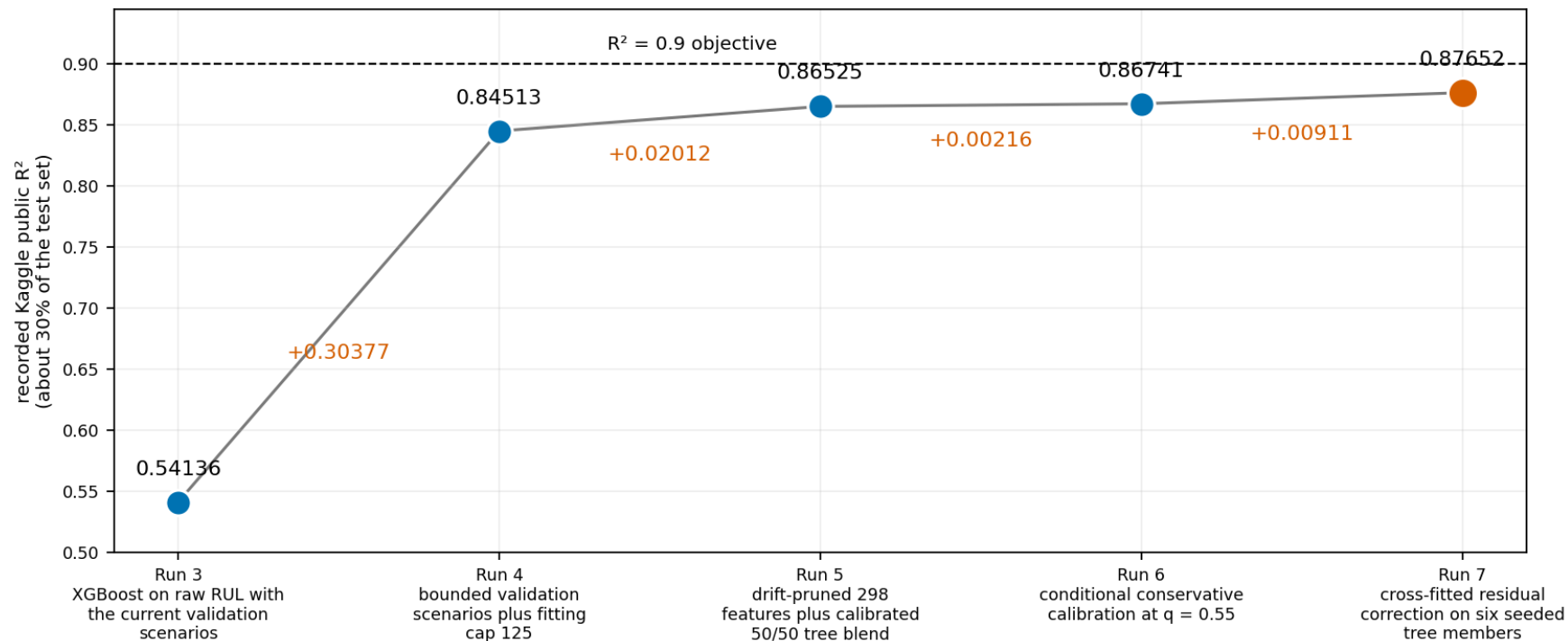


Figure: generated from kaggle_scores.csv and pipeline_experiments.md · claims C05-C12

Phase 0 decided which channels may carry information

Consistency inside a UAV, not pooled correlation, is what generalises to an unseen UAV

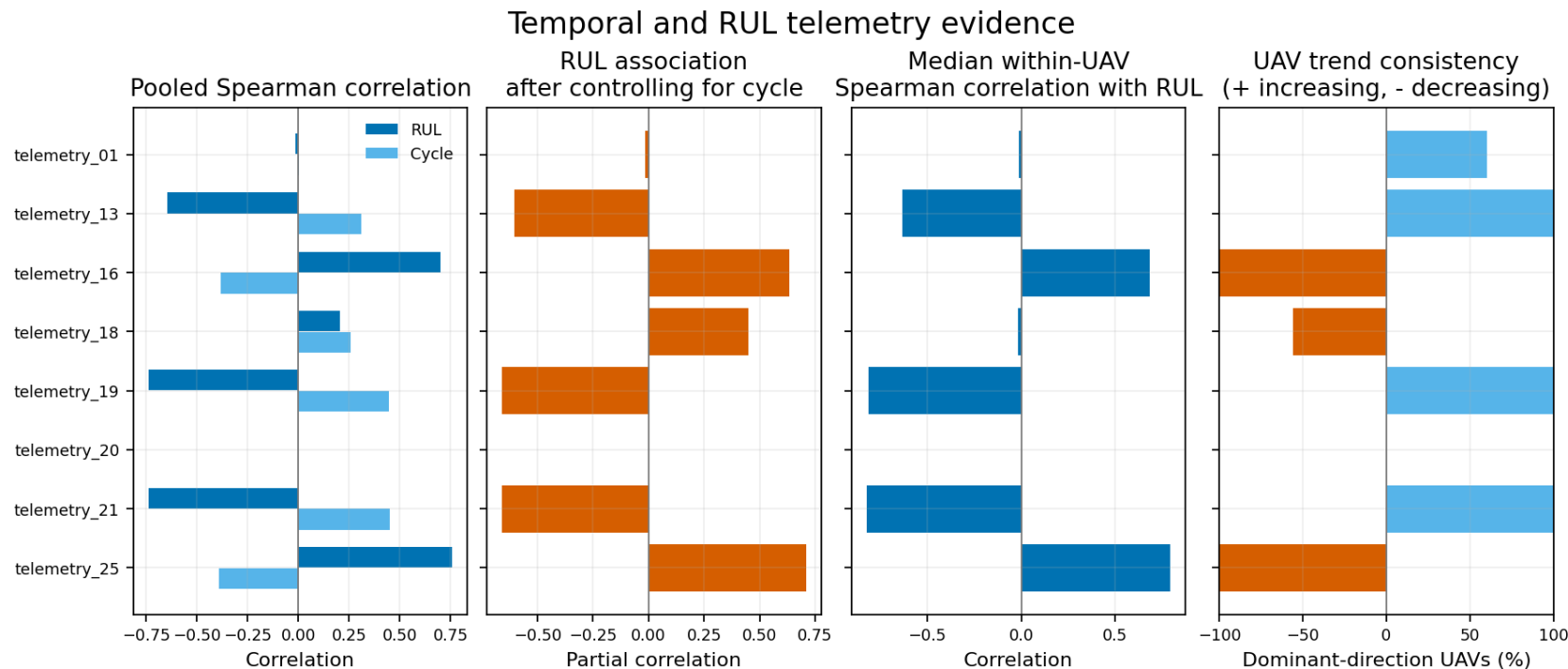


Figure: repository Phase 0 script re-run for the eight channels shown · claims C13-C19

Phase 1 decided what every later number is measured against

One hundred UAVs, not 24,720 rows: the UAV is the independent unit

Nested whole-UAV design, all five outer folds



fold 1 held out; folds 2-5 are the 80 training UAVs

Inside those 80: four inner folds select everything



features, hyperparameters, early stopping, blend weights

Rules that follow

- Every prefix of one UAV stays in one fold.
- Features use only cycles at or before the cutoff.
- Scalers refitted inside each training partition.
- Each UAV carries total training weight 1.
- Uncertainty resamples whole UAVs, never rows.

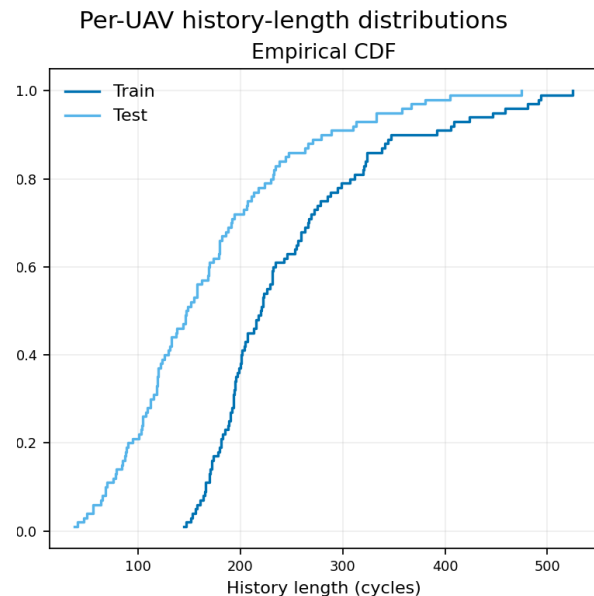


Figure: empirical-CDF panel of the repository Phase 0 history-length figure · claims C20-C27

Decision 1: bounded scenarios and a fitting cap at 125

Run 3 → Run 4, +0.30377 public. Blue = ExtraTrees, orange = XGBoost, paired against the control

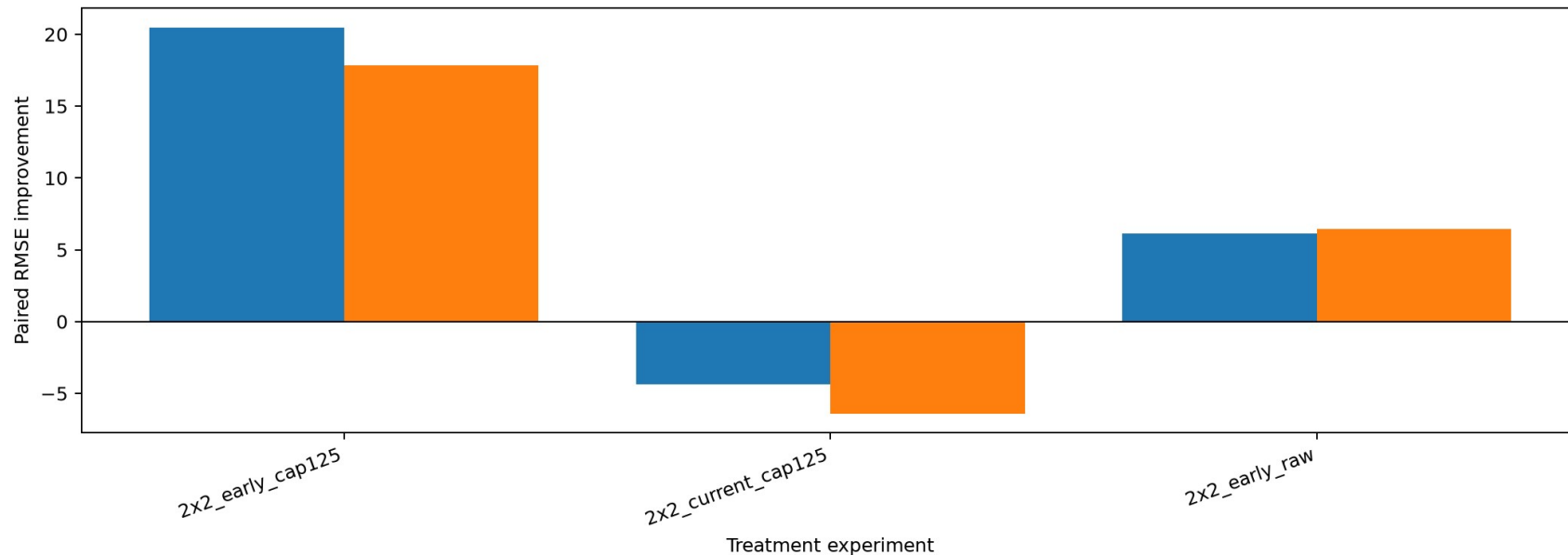


Figure: lower panel of the generated PE_2 paired comparison · claims C28-C35

Decision 2: which engineered features earn their place

Blue = ExtraTrees, orange = XGBoost, each treatment paired against the same control

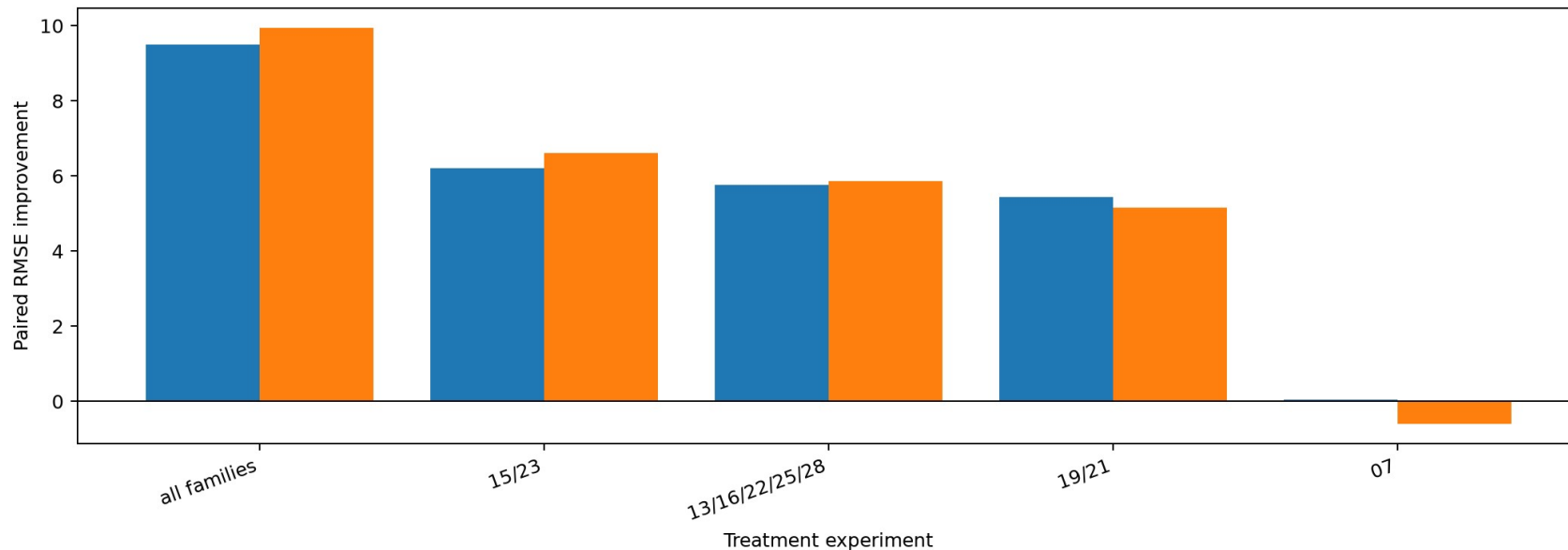


Figure: lower panel of the generated PE_2 paired comparison · claims C36-C42

Decision 3: the model family, chosen under one locked protocol

Twenty locked scenarios, 100 unseen UAVs, three seeds; no result was seen before the protocol froze

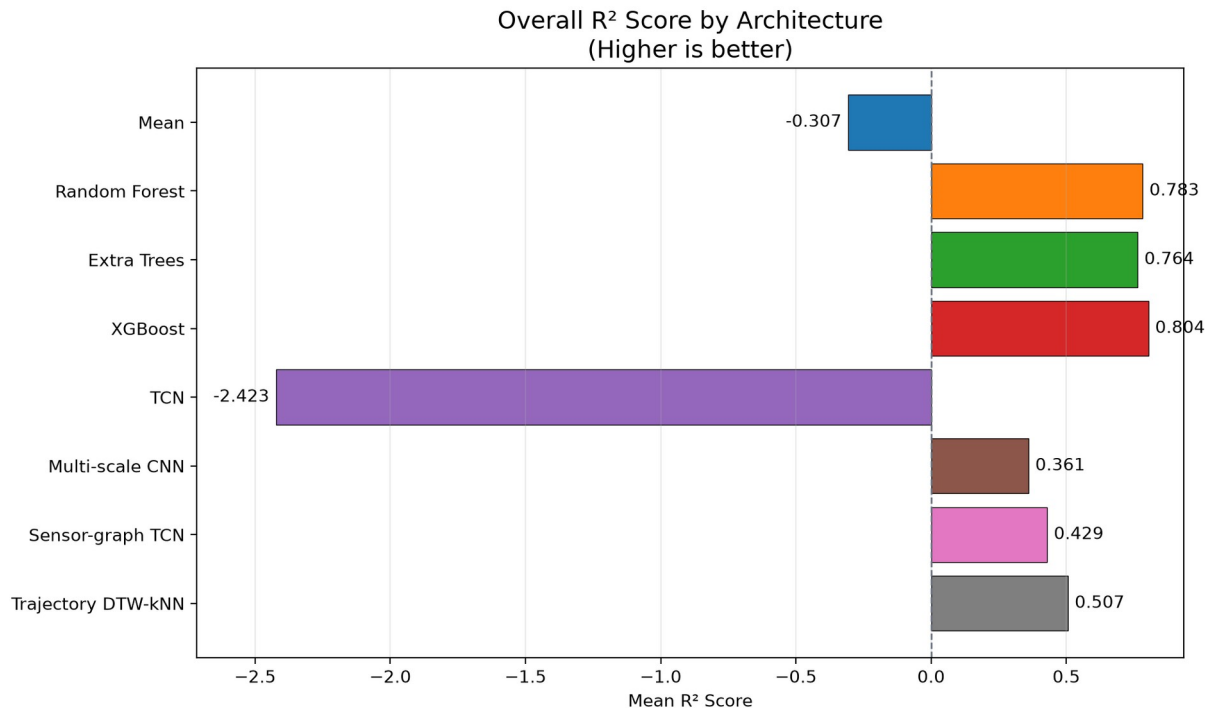
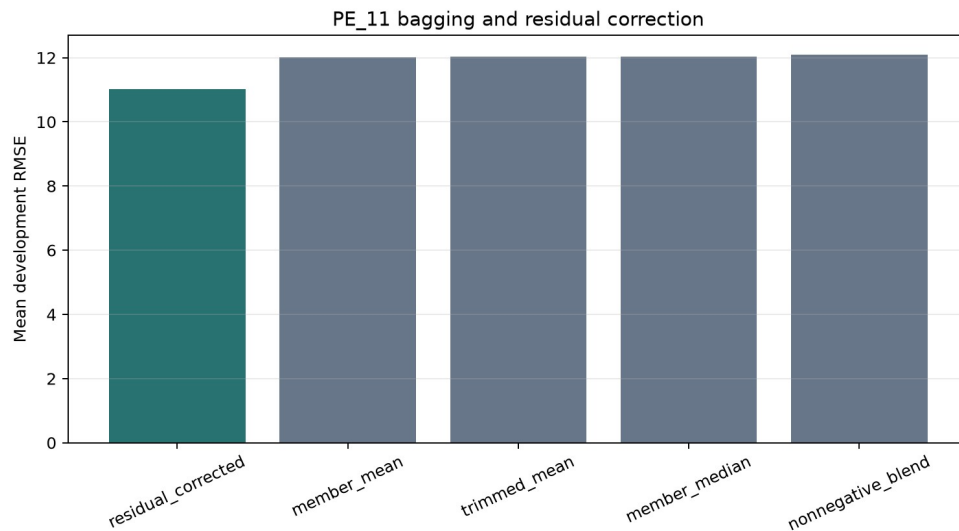
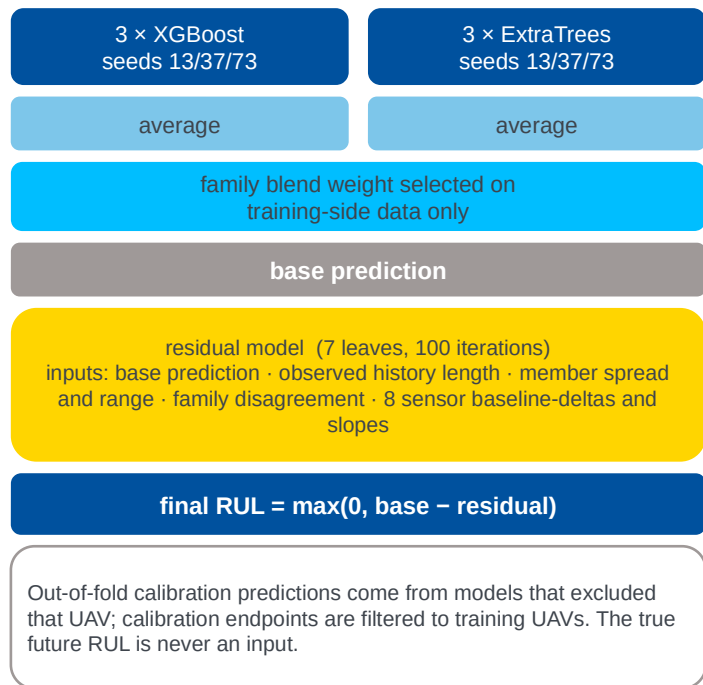


Figure: repository architecture study run 5 comparison, unchanged · claims C43-C53

Decision 4: a cross-fitted residual correction

Run 6 → Run 7, +0.00911 public. Averaging six members did not help; learning their residual did



11.5080 → 11.0243 cycles, 4.20% better, 5 of 5 fold wins. Refitted by the Phase 3 Run 7 adapter on all 100 training UAVs.

Figure: repository PE_11 comparison, unchanged · claims C54-C60

Decision 5: conditional conservative calibration

Run 5 → Run 6, +0.00216 public. A monotone trade-off: safety is bought with accuracy

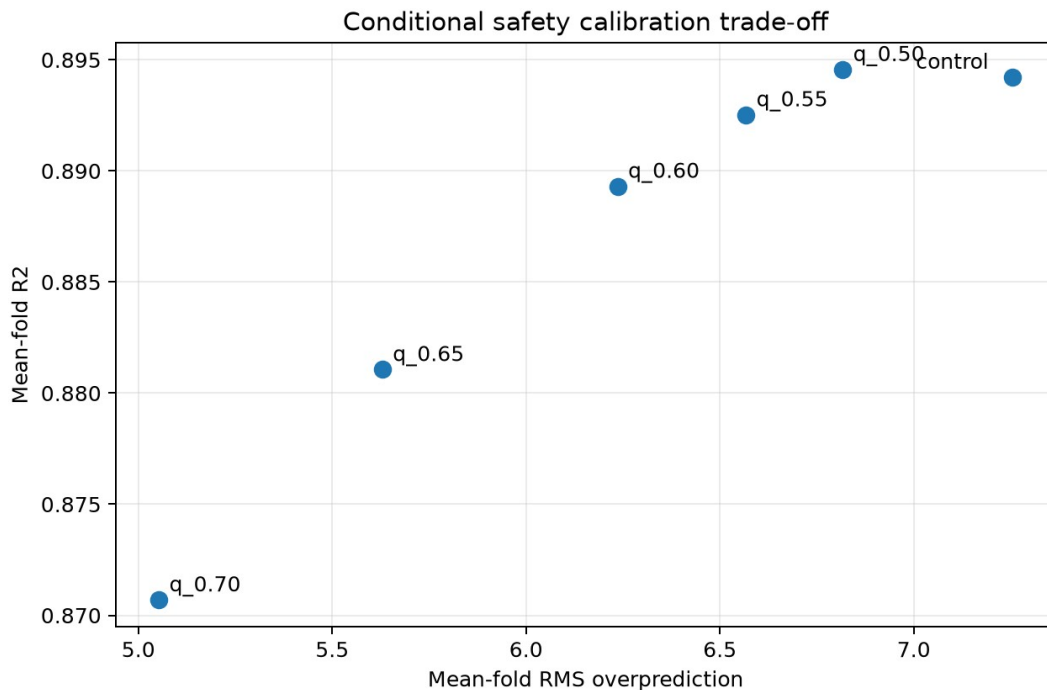


Figure: repository PE_4 trade-off, unchanged · claims C61-C65

What Run 7 still gets wrong

Development out-of-fold predictions: 500 rows, 100 UAVs, 444 distinct endpoints

Residual Corrected Tree Ensemble development OOF prediction alignment
the six-cycle offset is diagnostic and is not selected from this figure

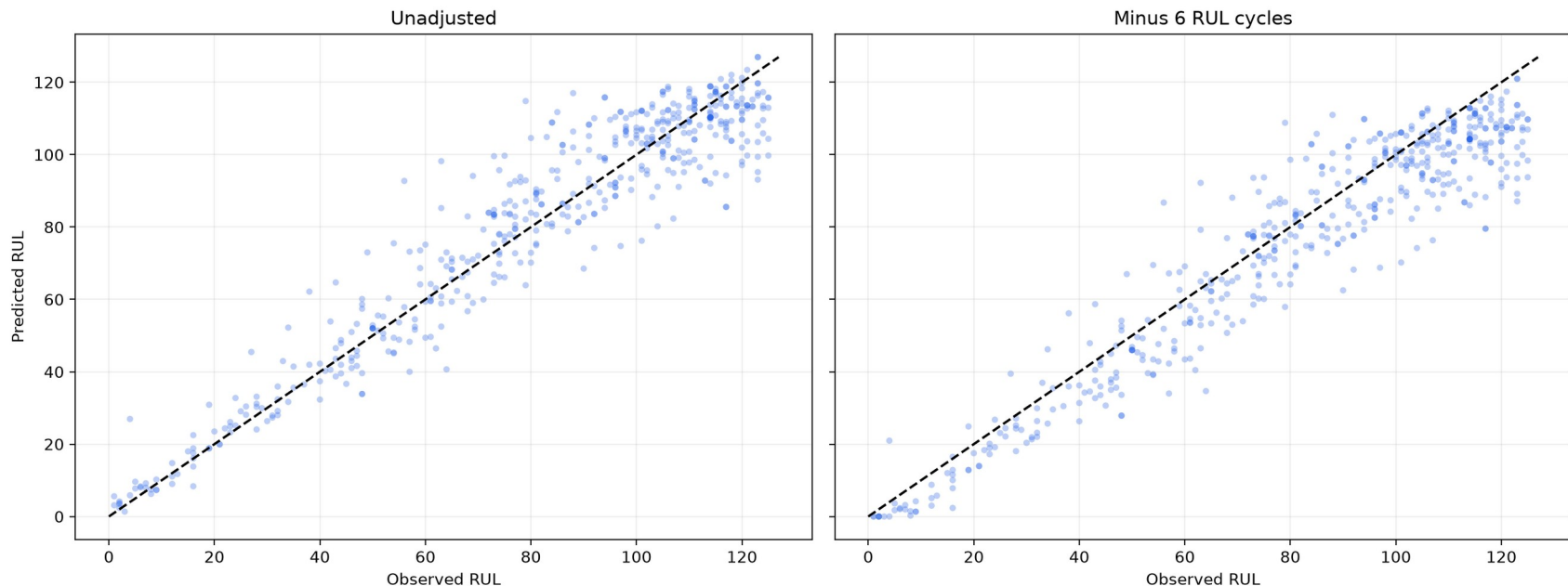
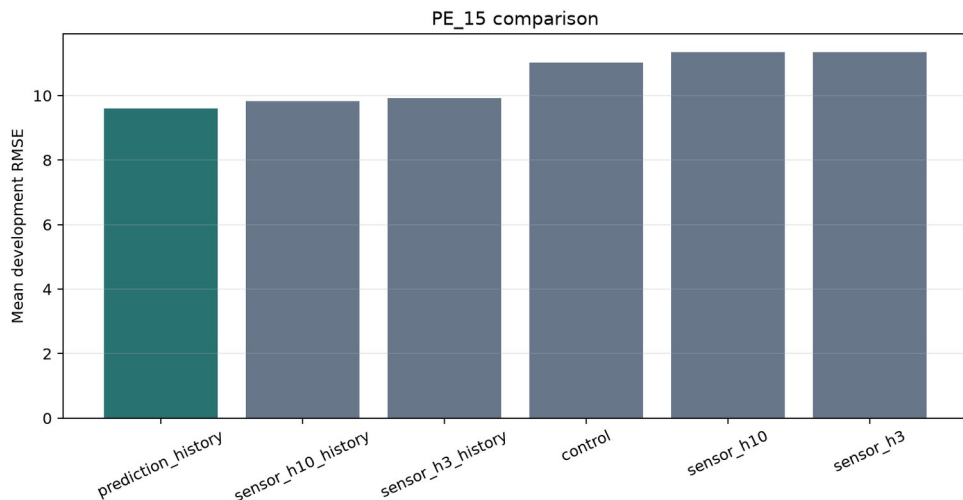


Figure: repository Phase 3 Run 7 report, unchanged · claims C66-C73

Why the chain stops at 0.87652

The strongest post-Run-7 candidate lost its entire gain under a complete nested refit



After Run 7: six candidates, all gated

candidate	mean RMSE	wins	outcome
PE_20 nested refit	+1.56% worse	2/5	rejected
PE_18 TabPFN blend	-1.30%	4/5	to confirm
PE_21 two new seeds	-1.12%	6/10	rejected
PE_24 regime gate	-1.90%	11/15	rejected
PE_26 short history	-0.39%	6/15	rejected
PE_27 UAV subsets	screen only	3/5	stopped

Every promotion gate is declared before the run. Nothing passed, so nothing was submitted, so the public score is still 0.87652.

PE_15 screen: 11.02 → 9.59 cycles, -13.0%, 5 of 5 folds

Figure: repository PE_15 comparison, unchanged; later results as numbers · claims C74-C84

Development performance and public performance

Development R^2 above 0.9; the recorded public score is 0.87652

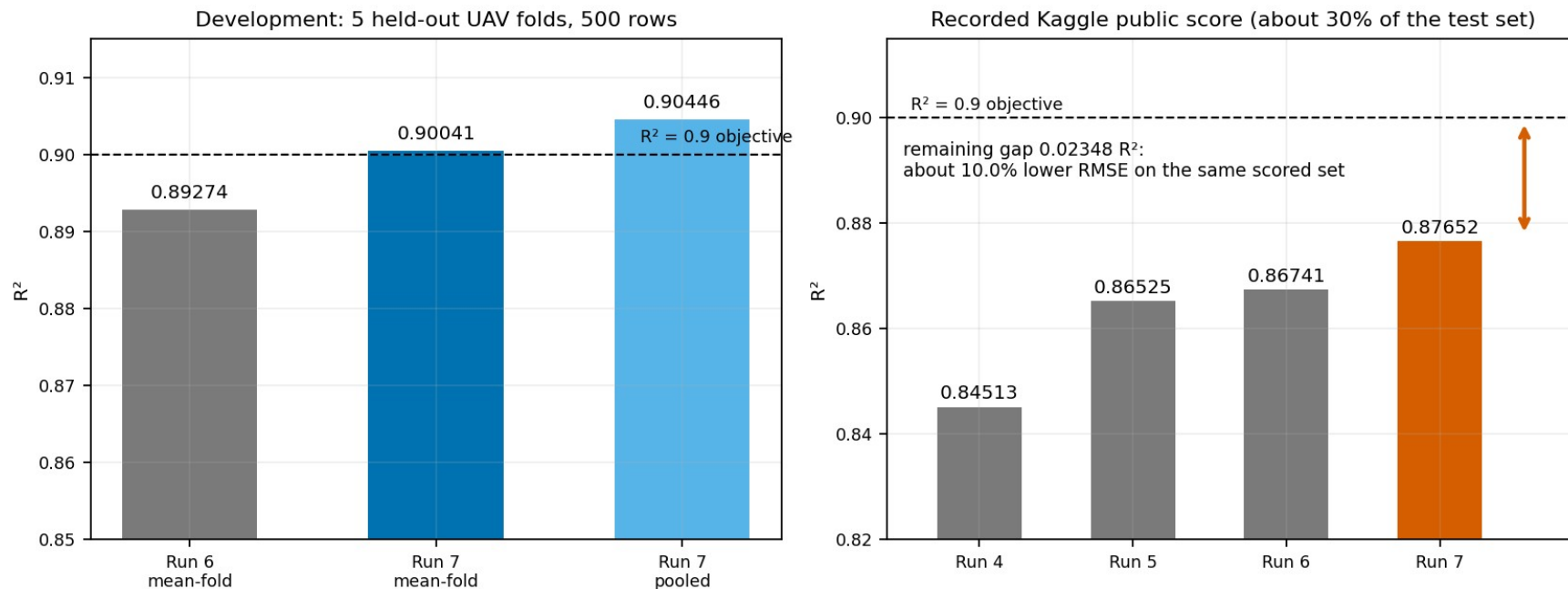


Figure: generated from run_6/run_7 report_summary.json and kaggle_scores.csv · claims C85-C93

What the next comparison can resolve

PE_28: ten feature representations × two model recipes — training started, no winner yet

feature set	features	what it varies
A_current	298	current pipeline representation (control)
B_other	266	alternative script: 22 sensors, simpler summaries
C_other_14	170	alternative summaries on 14 sensors
D_long_windows	266	longer trailing windows
E_compact_22	134	compact summaries, 22 sensors
F_baseline_10	266	10-cycle baseline averaging
G_all_windows	310	all window lengths retained
H_compact_14	86	compact summaries, 14 sensors
I_four_sensors	50	four sensors only
J_recent_slopes	310	added recent-slope features

Crossed with two model recipes

- the retained Run 7 ensemble
 - an XGBoost / CatBoost blend
- 100 screening evaluations, seed 20270107

Gates, declared in advance

- screen: 2% mean-RMSE gain, 4/5 folds
 - confirm: 2% gain, 8/10 folds, pooled $R^2 \geq 0.90$
- and a bootstrap interval wholly below zero

Status on 8 September 2026

PENDING — training started, no complete winner manifest.

Motivated by a user-reported public score of 0.87877 for a separate script (266 features, 22 sensors). That is not Run 7's score and not a matched experiment.

Source: PE_28 README.md and run_1 input_verification.json, inspected 8 September 2026 · claims C94-C99

How we reached 0.87652

One decision did most of the work; four smaller ones did the rest

phase	decision	submissions	public R ² change
Phase 1 + 2	Bounded validation scenarios with a fitting cap at 125	Run 3 → 4	+0.30377
Phase 0 + 2	Drift-pruned 298 features, calibrated 50/50 tree blend	Run 4 → 5	+0.02012
Phase 2	Conditional conservative calibration at $q = 0.55$	Run 5 → 6	+0.00216
Phase 2	Cross-fitted residual correction on six seeded members	Run 6 → 7	+0.00911

Each row is a submission-to-submission difference, not an isolated ablation: a run can carry more than one change. Evaluation and target design dominated model choice.

What we do not claim

Development mean-fold R^2 is 0.90041; the recorded public score is 0.87652. Closing that needs about 10.0% lower RMSE on the same scored set, and no candidate after Run 7 passed a nested confirmation.

Source: evidence_ledger.csv · claims C100

Not part of the 20-minute talk. Held for questions.

Backup slides

Backup: full locked architecture comparison

Architecture study run 5, settings version 11; 20 locked scenarios × 100 UAVs

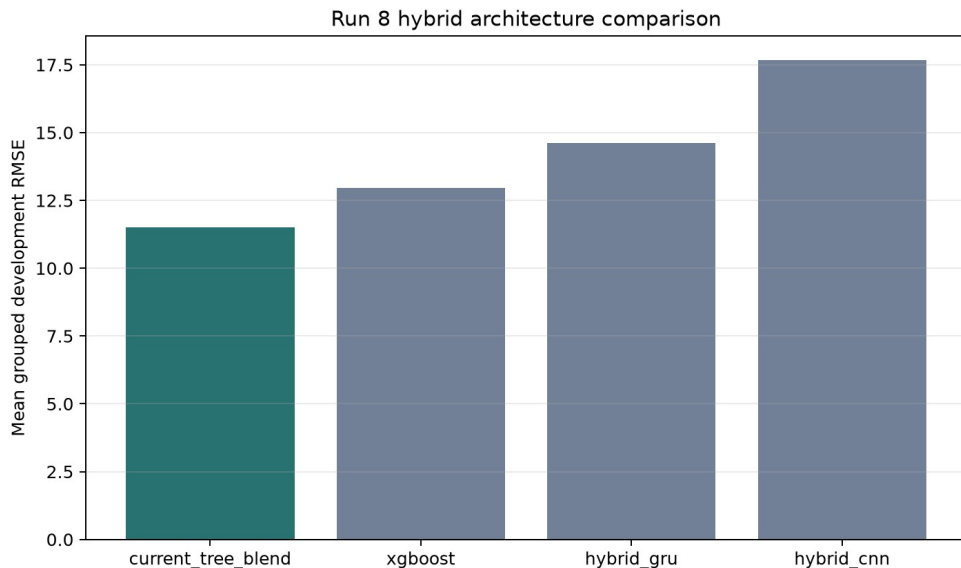
architecture	seeds	R ² mean	RMSE mean	R ² seed SD	R ² 95% UAV-bootstrap interval
XGBoost	3	0.8041	28.4989	0.0055	0.7528 - 0.8437
Random Forest	3	0.7827	30.0138	0.0030	0.7058 - 0.8348
Extra Trees	3	0.7640	31.2779	0.0027	0.6671 - 0.8372
Trajectory DTW-kNN	1	0.5067	45.2239	0	0.3804 - 0.6106
Sensor-graph TCN	3	0.4294	48.2278	0.1487	0.1602 - 0.6359
Multi-scale CNN	3	0.3607	51.4209	0.0622	0.2317 - 0.4691
Mean baseline	1	-0.3066	73.6023	0	-0.4848 - -0.1758
TCN	3	-2.4231	111.2088	2.1691	-9.7147 - 0.6382

Metrics are averaged across seeds; predictions are not averaged, because that would create an undeclared ensemble. Intervals are 1,000 whole-UAV bootstrap resamples of fixed predictions.

Source: runs/run_5/7_architecture_comparison/architecture_comparison.csv

Backup: sequence and hybrid models, in full

The matched rematch under the current protocol, and the dedicated temporal study



Architecture study run 8, early/middle scenarios and cap-125 target. Neither hybrid won a single fold out of five.

Figure: repository architecture study run 8, unchanged; run 7 temporal summary

Dedicated temporal study (run 7)

model	mean R ²	mean RMSE
LSTM	0.6531	19.570
GRU	0.6424	19.832
TCN	0.5969	21.084
Multi-scale CNN	0.5325	22.712

Its own frozen sampling policy and lookback 20, so these values are not on a common scale with the left panel. The gate it had to clear was about R^2 0.89 and RMSE 10.7.

State the finding for the architectures, representations and budgets tested here — not as a claim about deep learning and RUL in general.

Backup: the complete cap and scenario matrix

Why the two scenario profiles cannot share one R^2 axis

scenarios	fitting target	model	inner RMSE	fold SD	inner R^2	bias	overpred. rate
current	raw	XGBoost	29.8059	1.7513	0.7708	+5.2165	67.35%
current	raw	ExtraTrees	32.7578	2.6545	0.7264	+4.4906	66.00%
current	cap 125	XGBoost	36.2141	2.5645	0.6690	-10.8488	46.70%
current	cap 125	ExtraTrees	37.1100	2.5884	0.6525	-10.7370	50.95%
early/middle	raw	XGBoost	23.3667	1.5326	0.4867	+11.4973	71.35%
early/middle	raw	ExtraTrees	26.6406	2.9202	0.2703	+10.6021	67.60%
early/middle	cap 125	XGBoost	11.9746	0.9418	0.8661	-0.9007	47.30%
early/middle	cap 125	ExtraTrees	12.2693	0.8427	0.8598	-1.0864	47.15%

The early/middle profile assigns test-like cutoffs only where true RUL lies between 1 and 125. Matching the cutoff marginal does not make the joint distribution match the hidden test endpoints, and restricting the target support changes the variance in the R^2 denominator. The fitting cap of 125 is a separate mechanism from evaluation-target support and from prediction clipping; metrics always use raw endpoint labels, and the only clipping applied is a nonnegative floor.

Source: pipeline_experiments.md; _internal/shared_settings.toml

Backup: residual correction, nested calibration and the safety table

How the Run 7 adapter avoids leaking, and what the calibration bought

What the Run 7 adapter does

- Splits training UAVs internally into four groups.
- Fits the six tree members without the held group.
- Predicts it; repeats, so no UAV is seen by its own model.
- Filters calibration endpoints to training UAVs and balances weight per UAV.
- Fits the residual head there, then predicts the external held-out UAVs once.
- The earlier PE_11 report is not a fully nested estimate of its own stack; the Run 7 adapter does not share that dependency.
- Its remaining limitation: configurations and the workflow were chosen on development data now examined many times.

Conditional calibration policies (PE_4)

policy	mean-fold R ²	RMSE	RMS overpred.
control	0.89422	10.5306	7.255
q = 0.50	0.89456	10.5354	6.817
q = 0.55	0.89253	10.6430	6.567
q = 0.60	0.88931	10.8144	6.236
q = 0.65	0.88106	11.2152	5.629
q = 0.70	0.87070	11.6989	5.053

Rule declared in advance: only policies within 0.005 mean-fold R² of the best are eligible; among those, minimise RMS overprediction, then overprediction rate, then RMSE.

A later uncertainty-scaled variant improved near-failure RMS overprediction by 11.31% but raised RMSE by 1.71% and was rejected.

Source: residual_ensemble_contract.json; PE_4 calibration_summary.csv; r2 research report

Backup: representation experiments that were rejected

Seven matched comparisons, all development-only, all with their control retained

representation tested	matched development result	decision
Signal compression (median / PCA indices)	R^2 fell by roughly 0.09-0.11 in both families	keep individual features
Dense training prefixes (stride 5)	R^2 -0.0338 XGBoost, -0.0362 ExtraTrees; lost every fold	keep 20 test-like cutoffs
Per-UAV robust normalisation	combined set: +0.0051 XGBoost, -0.0065 ExtraTrees	keep raw features
Failure-cycle target instead of RUL	ΔR^2 -0.4002 XGBoost, -0.5073 ExtraTrees; 0/5 folds	keep RUL target
Personalised degradation-onset targets	XGBoost RMSE 12.15 $\rightarrow$ 19.59-19.69	keep cap 125
Drift-feature pruning	domain AUC 0.8565, but no treatment passed the accuracy gate	keep all features
Population degradation features	mean RMSE +2.5% (HGB) and +11.8% (ridge)	keep control

PE_28 now revisits the representation question deliberately: ten feature sets from 50 to 310 features crossed with two model recipes, varying sensor coverage, summary complexity, baseline averaging and window length one factor at a time. Screening uses grouped seed 20270107; confirmation uses seeds 20270117 and 20270127 on the same UAV population, so it is a robustness check rather than an untouched holdout.

Source: pipeline_experiments.md experiment register

Backup: uncertainty, endpoint duplication and score provenance

What the bootstrap intervals do and do not cover

What the bootstrap covers

- 1,000-3,000 resamples of whole UAVs, never of rows.
- Run 7 pooled R^2 interval 0.87779 - 0.92772.
- Run 7 minus Run 6 RMSE interval -1.252 to +0.446 cycles.
- It resamples fixed saved predictions, so it excludes training and model-selection uncertainty.
- A positive mean with an interval crossing zero is uncertain evidence, not proof of no effect.

Repeated endpoints

- 500 development rows contain 444 distinct (UAV, cutoff) pairs.
- Removing duplicates changes pooled R^2 only from 0.90446 to 0.90494.
- Duplication therefore does not explain the headline number, but it does affect weighting and precision.
- The UAV, not the endpoint, stays the resampling unit.

Public-score provenance

- Public scores were transcribed from user-supplied Kaggle screenshots recorded on 7 September 2026, not from a live lookup.
- The public leaderboard uses about 30% of the test data; the final standing uses the other 70%.
- The 0.87877 figure for the alternative script is user-reported and has no submission artifact in this repository.

Source: diagnostic_summary.json; kaggle_scores.csv; PE_14 notes in the r2 report